The Theory of Consumer Choice

EC 111 Principles of Microeconomics

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Based on Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 22. Copyright Cengage.

In this chapter

- How does the budget constraint represent the choices a consumer can afford?
- How do indifference curves represent the consumer's preferences?
- What determines how a consumer divides her resources between two goods?
- How does the theory of consumer choice explain decisions such as how much a consumer saves, or how much labor she supplies?

Part 1

The Budget Constraint: What a Consumer Can Afford

The budget constraint

Trade-offs:

- Buying more of one good leaves less income to buy other goods.
- Working more hours means more income and more consumption, but less leisure time.
- Reducing saving now allows more consumption today but reduces future consumption.

Budget constraint. The limit on the consumption bundles that a consumer can afford.

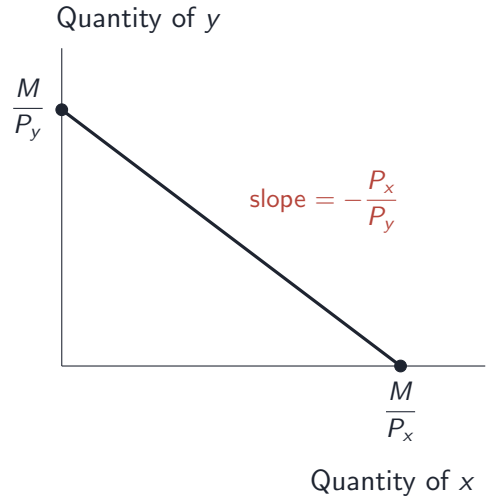
The budget constraint as an equation

P_x and P_y are the prices of goods x and y.
M is money income.

$$P_x\,x + P_y\,y = M$$

$$y = \frac{M}{P_y} - \frac{P_x}{P_y}\,x$$

Line 1: spending equals income.
Line 2: subtract $P_x x$ from both sides, then divide by P_y.



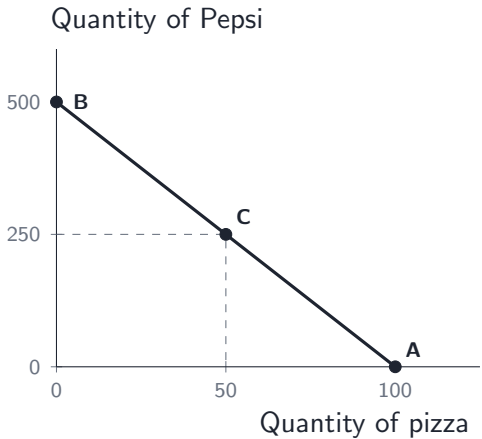
The example: pizza and Pepsi

Mankiw's consumer buys two goods, pizza and Pepsi.

- Income: \$1,000 a month, all of it spent.
- A pizza costs \$10. A liter of Pepsi costs \$2.

Pizzas	Liters of Pepsi	Total spending
100	0	\$1,000
50	250	\$1,000
0	500	\$1,000

Every bundle on the line costs exactly \$1,000.



The equation with our numbers

Put Mankiw's numbers into the equation, with pizza as x and Pepsi as y:

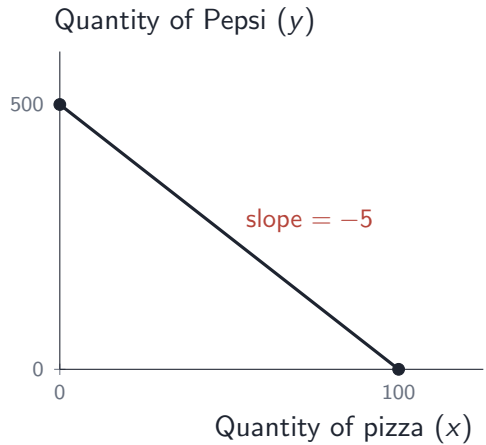
$$M = \$1{,}000, \qquad P_x = \$10, \qquad P_y = \$2.$$

$$y \text{ axis:} \quad \frac{M}{P_y} = \frac{1{,}000}{2} = 500$$

$$x \text{ axis:} \quad \frac{M}{P_x} = \frac{1{,}000}{10} = 100$$

$$\text{slope:} \quad -\frac{P_x}{P_y} = -\frac{10}{2} = -5$$

So the budget constraint is $y = 500 - 5x$.



The slope is the relative price

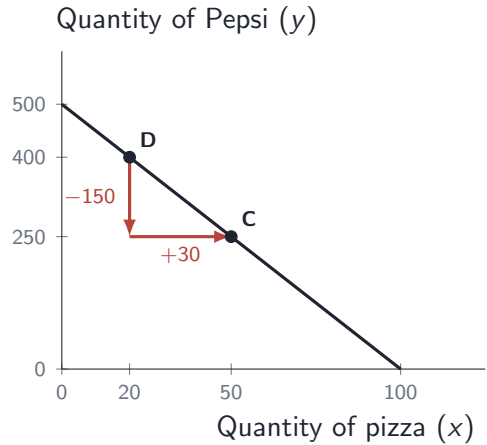
Move along the line from D to C:

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{250 - 400}{50 - 20} = \frac{-150}{30} = -5$$

Result. The slope of the budget constraint is the relative price, P_x/P_y.

Why. A pizza costs \$10 and a liter of Pepsi \$2. Buying one more pizza means giving up 5 liters.

Finding. The slope tells you what one more unit of x costs, measured in units of y.



What moves the budget constraint: income

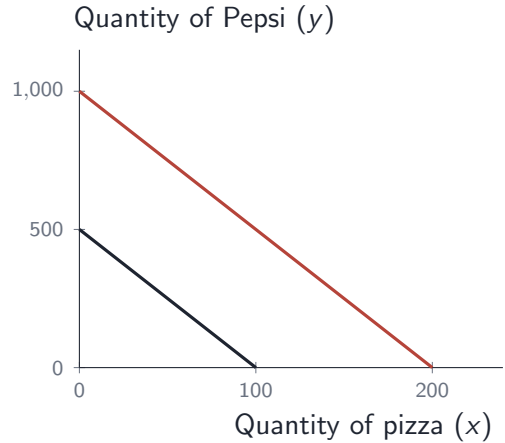
Income doubles, from \$1,000 to \$2,000.
Prices do not change.

$$y \text{ axis:} \quad \frac{M}{P_y} = \frac{2{,}000}{2} = 1{,}000$$

$$x \text{ axis:} \quad \frac{M}{P_x} = \frac{2{,}000}{10} = 200$$

$$\text{slope:} \quad -\frac{P_x}{P_y} = -5 \quad \text{(unchanged)}$$

Finding. M appears only in the end points. A change in income moves the line out in parallel.



Black: initial. Red: new.

What moves the budget constraint: a price

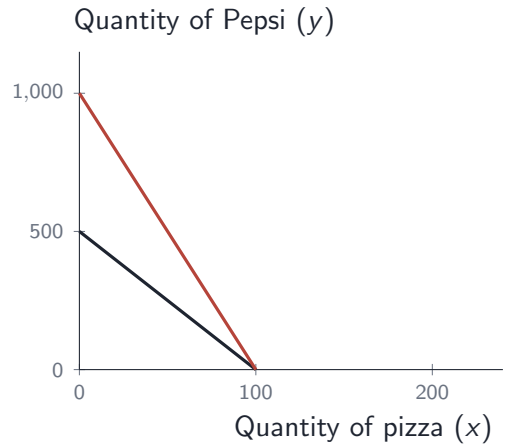
The price of Pepsi halves, from \$2 to \$1. Income and P_x do not change.

$$y \text{ axis:} \quad \frac{M}{P_y} = \frac{1{,}000}{1} = 1{,}000$$

$$x \text{ axis:} \quad \frac{M}{P_x} = 100 \quad \text{(unchanged)}$$

$$\text{slope:} \quad -\frac{P_x}{P_y} = -\frac{10}{1} = -10$$

Finding. A price appears in one end point and in the slope. The line pivots around the end point that does not change.



Black: initial. Red: new.

What moves the budget constraint: the other price

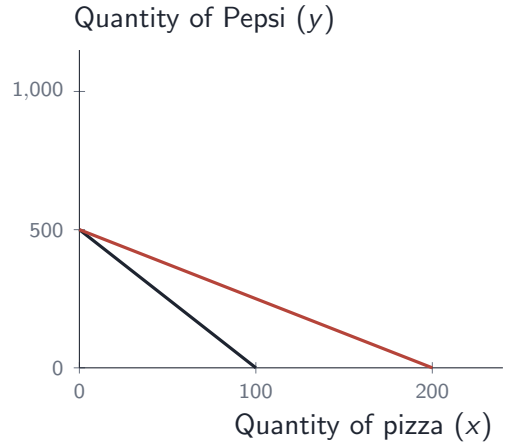
The price of pizza halves, from \$10 to \$5. Income and P_y do not change.

$$y \text{ axis:} \quad \frac{M}{P_y} = 500 \quad \text{(unchanged)}$$

$$x \text{ axis:} \quad \frac{M}{P_x} = \frac{1{,}000}{5} = 200$$

$$\text{slope:} \quad -\frac{P_x}{P_y} = -\frac{5}{2} = -2.5$$

Finding. Now the pizza end point moves and the line pivots around the Pepsi end point. It becomes flatter: a pizza costs only 2.5 liters.



Black: initial. Red: new.

What moves the budget constraint: summary

Read every case off one equation:

$$y = \frac{M}{P_y} - \frac{P_x}{P_y}\,x$$

Change	End points	Slope
Income M rises	both move out	unchanged: parallel shift
P_y falls	y end point moves out	steeper
P_x falls	x end point moves out	flatter

Income moves the line. A relative price change rotates it.

Part 2

Preferences: What a Consumer Wants

Utility and marginal utility

Utility. A measure of the happiness or satisfaction a consumer gets from a bundle of goods.

Marginal utility. The increase in utility from one more unit of a good. It is the slope of the total-utility curve: $MU = \Delta U / \Delta Q$, where Δ means "change in" and Q is the quantity of the good.

Diminishing marginal utility. The more of a good the consumer already has, the less utility one more unit adds.

Total utility still rises with each unit, but by less and less: the total-utility curve gets flatter.

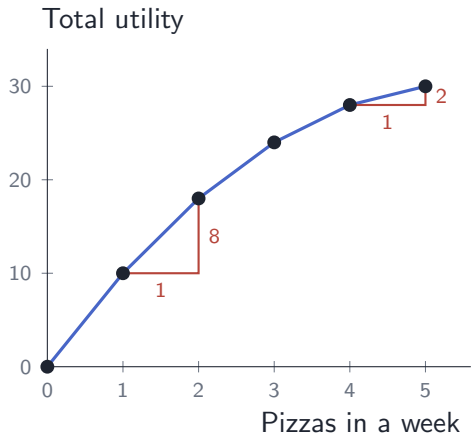
Diminishing marginal utility: an example

Her utility from pizzas in a week:

Pizzas	Total utility	$MU = \Delta U / \Delta Q$
1	10	$10/1 = 10$
2	18	$8/1 = 8$
3	24	$6/1 = 6$
4	28	$4/1 = 4$
5	30	$2/1 = 2$

From 1 to 2 pizzas, utility rises from 10 to 18: $\Delta U = 8$ for $\Delta Q = 1$, so the slope is 8. From 4 to 5 pizzas the slope is only 2.

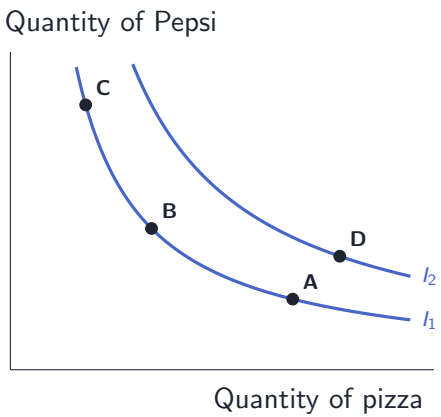
Finding. Marginal utility is the slope of the total-utility curve. The slope falls as she has more pizza.



Indifference curves

Indifference curve. A curve that shows the bundles that give the consumer the same level of satisfaction.

- A, B and C are on the same curve, I_1: she is equally happy with each.
- D is on a higher curve, I_2: she prefers D to every bundle on I_1.
- D beats A: it has more of both goods.
- D beats C, even with less Pepsi: the extra pizza more than makes up for it.



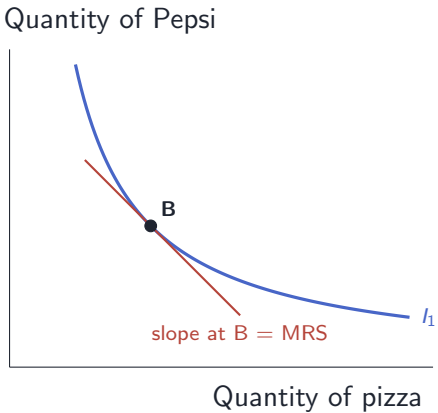
The marginal rate of substitution

Marginal rate of substitution (MRS). The rate at which the consumer is willing to trade one good for the other.

Result. The MRS at B is the slope of the line that just touches the curve at B.

Why. Along the curve she is equally happy. So the slope at B tells us how much extra Pepsi she needs to make up for one less pizza, starting from B.

Finding. The MRS is the slope of the indifference curve at that point (ignoring the minus sign). Each point has its own slope.



Four properties of indifference curves

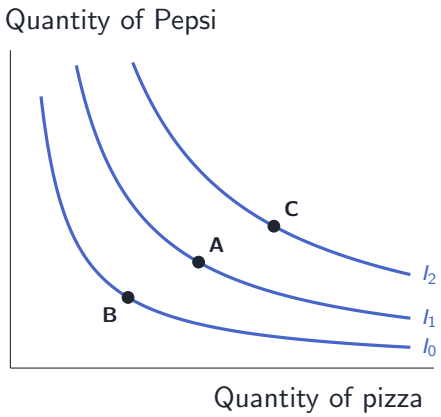
1. **Higher indifference curves are preferred to lower ones.** People usually prefer to consume more rather than less.
2. **Indifference curves slope downward.** If the quantity of one good falls, the quantity of the other must rise for the consumer to be equally happy.
3. **Indifference curves do not cross.** If they crossed, a bundle with more of both goods would be no better than one with less.
4. **Indifference curves are bowed inward.** People are more willing to trade away goods they have in abundance: the MRS falls along the curve.

Property 1: higher indifference curves are preferred

Result. She prefers every bundle on I_2 to every bundle on I_1, and every bundle on I_1 to every bundle on I_0.

Why. Start from A on I_1. Moving out to C gives her more of both goods. Moving in to B gives her less of both.

Finding. Further from the origin means more of both goods, so higher satisfaction.

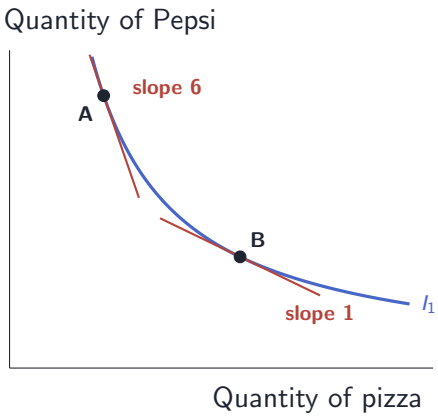


Properties 2 and 4: downward-sloping and bowed inward

Result. The curve slopes down, and it gets flatter as she moves down it: the MRS falls from 6 at A to 1 at B.

Why. Preferences are convex: she is more willing to give up a good she has in abundance. At A she has much Pepsi and little pizza, so she gives up 6 liters to get one more pizza. At B she has little Pepsi, so she gives up only 1.

Finding. A falling MRS is what makes indifference curves bowed inward.



Property 3: indifference curves cannot cross

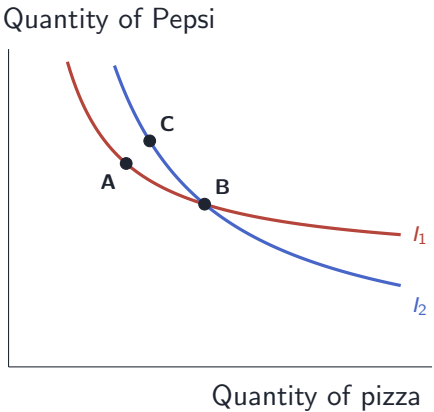
Result. Two indifference curves cannot cross.

Why. Suppose they did.

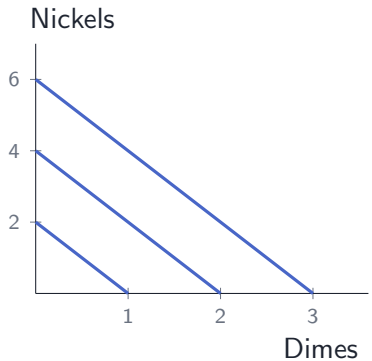
- A and B are on I_1: A is as good as B.
- B and C are on I_2: B is as good as C.
- So A is as good as C.
- But C has more of both goods than A, so C is better.

A cannot be both as good as C and worse than C.

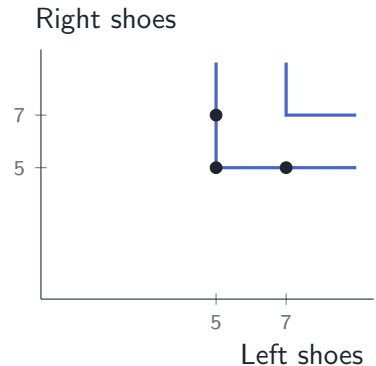
Finding. Crossing curves would contradict "more is better".



Two extreme cases

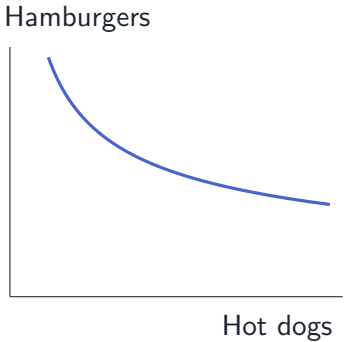


Perfect substitutes. Straight lines. 2 nickels always trade for 1 dime: the MRS is fixed at 2.



Perfect complements. Right angles. 5 left and 7 right shoes make only 5 pairs.

Between the extremes: close substitutes and close complements



Close substitutes: only slightly bowed. She swaps one for the other at a fairly steady rate.



Close complements: very bowed. Extra socks without shoes add little.

Finding. The easier the goods are to substitute, the less bowed the curve.

Part 3

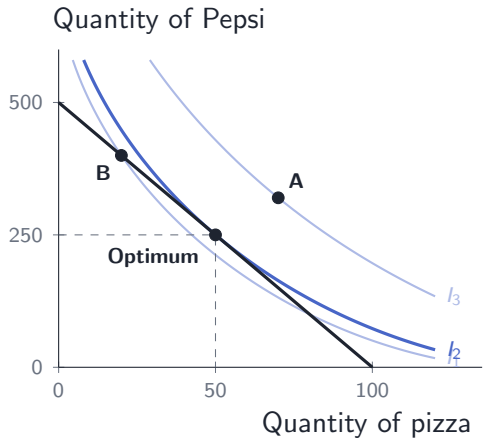
Optimization: What a Consumer Chooses

The consumer's optimal choice

She wants the highest indifference curve she can reach. Her budget limits her to bundles on or below the budget constraint.

- A is on I_3, but it costs \$1,340: she cannot afford it.
- B is affordable, but it is on the low curve I_1.
- The optimum is the bundle on the budget line that reaches the highest curve, I_2: 50 pizzas and 250 liters.

Optimum. The best bundle the consumer can afford.



At the optimum, MRS = relative price

x = pizza, y = Pepsi

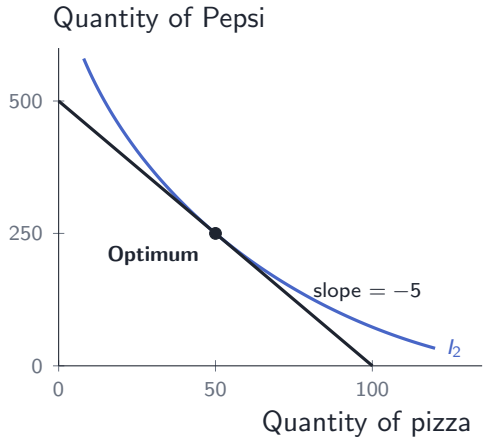
At the optimum the indifference curve just touches the budget constraint, so the two have the same slope.

Budget constraint: $\dfrac{P_x}{P_y} = \dfrac{\$10}{\$2} = 5$

Indifference curve: $\text{MRS} = 5$

Result. The rate at which she is willing to trade (the MRS) equals the rate at which the market lets her trade (the relative price).

Finding. At the optimum, $\text{MRS} = P_x/P_y$.



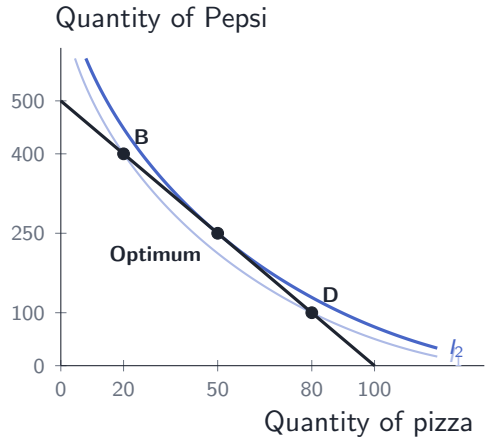
Why the optimum must have MRS = relative price

At B the MRS is about 8.5. She would give up 8.5 liters to get one more pizza. The market asks only 5. So she buys one more pizza, pays 5 liters, and is better off.

At D the MRS is about 2.9. She would give up only 2.9 liters for a pizza, but one pizza sells for 5 liters. So she buys one less pizza, gets 5 liters, and is better off.

Moving from B or D toward the optimum, the MRS moves toward 5. She stops when the two rates are equal.

Finding. Any bundle where MRS $\neq P_x/P_y$ can be improved on.



Utility: why the MRS equals MU_x/MU_y

x = pizza, y = Pepsi

Move along one indifference curve: Δx more pizza and Δy of Pepsi, where Δy is negative because she gives Pepsi up.

$$\Delta U = \underbrace{MU_x\,\Delta x}_{\text{gain from pizza}} + \underbrace{MU_y\,\Delta y}_{\text{loss on Pepsi}} \qquad \text{change in utility}$$

$$0 = MU_x\,\Delta x + MU_y\,\Delta y \qquad \text{same curve, so utility does not change}$$

$$MU_x\,\Delta x = -MU_y\,\Delta y \qquad \text{subtract } MU_y\,\Delta y \text{ from both sides}$$

$$-\frac{\Delta y}{\Delta x} = \frac{MU_x}{MU_y} \qquad \text{divide both sides by } \Delta x \text{ and by } MU_y$$

$-\Delta y/\Delta x$ is the Pepsi she gives up per extra pizza: the MRS.

MRS $= MU_x/MU_y$

Two slopes, one rule

x = pizza, y = Pepsi

Both slopes, without their minus signs:

indifference curve (previous slide): $$-\frac{\Delta y}{\Delta x} = \frac{MU_x}{MU_y}$$

budget constraint (its equation): $$-\frac{\Delta y}{\Delta x} = \frac{P_x}{P_y}$$

At the optimum the two curves touch, so their slopes are equal:

$$\frac{MU_x}{MU_y} = \frac{P_x}{P_y} \quad \Rightarrow \quad \frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

multiply both sides by MU_y, then divide both sides by P_x

At the optimum, marginal utility per dollar is the same for both goods.

What the rule means

x = pizza, y = Pepsi

- One dollar buys $1/P_x$ of a pizza. So the last dollar spent on pizza adds $MU_x \times (1/P_x) = MU_x/P_x$ to her utility.
- In the same way, the last dollar spent on Pepsi adds MU_y/P_y.

At her optimum. Suppose one more pizza adds 10 units of utility and one more liter adds 2. Then MRS $= 10/2 = 5$, equal to the relative price.

pizza: $10/\$10 = 1$ per dollar Pepsi: $2/\$2 = 1$ per dollar

At the optimum, the last dollar spent on each good adds the same utility.

Why the rule must hold

x = pizza, y = Pepsi

Suppose that, at some bundle on her budget line, the last dollar spent on pizza adds 1.5 units of utility and the last dollar spent on Pepsi adds 0.8.

She moves one dollar from Pepsi to pizza. Her spending is unchanged. She gains 1.5 and loses 0.8, so her utility rises by 0.7. That bundle was not her best.

In symbols, the starting position was

$$\frac{MU_x}{P_x} > \frac{MU_y}{P_y} \quad \Longleftrightarrow \quad \frac{MU_x}{MU_y} > \frac{P_x}{P_y} \qquad \text{(multiply both sides by } P_x/MU_y\text{)}$$

The left side of the second inequality is her MRS. So this is the position at B two slides back: MRS above the relative price. As she shifts spending toward pizza, her MRS falls toward P_x/P_y and the gap closes.

She gains by reallocating until the last dollar on pizza and the last dollar on Pepsi add the same utility.

Two ways of saying the same thing

x = pizza, y = Pepsi

Indifference curves	**Utility**
She reaches the highest indifference curve she can afford.	She maximizes utility subject to her budget.
At the optimum the indifference curve is tangent to the budget constraint: MRS $= P_x/P_y$.	At the optimum marginal utility per dollar is the same for every good: $MU_x/P_x = MU_y/P_y$.

The two conditions are the same condition, written two ways.

How changes in income affect her choices

An increase in income shifts the budget constraint out in parallel: the relative price has not changed. She can reach a higher indifference curve.

Normal good. A good for which an increase in income raises the quantity demanded.

Inferior good. A good for which an increase in income reduces the quantity demanded.

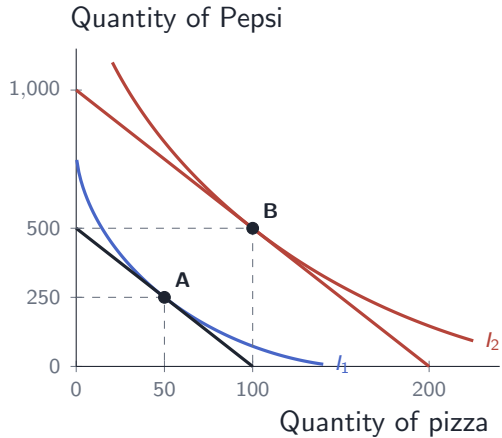
Most goods are normal. Bus rides are an example of an inferior good: as income rises, people are more likely to own cars or use Uber, and less likely to ride the bus.

An increase in income: normal goods

Her income rises from \$1,000 to \$2,000. Prices stay at \$10 and \$2.

- The budget line shifts out in parallel, from 100 and 500 to 200 and 1,000.
- Her optimum moves from A $(50, 250)$ to B $(100, 500)$, on the higher curve I_2.
- She buys more pizza and more Pepsi.

Finding. Both goods are normal here: more income, more of each.

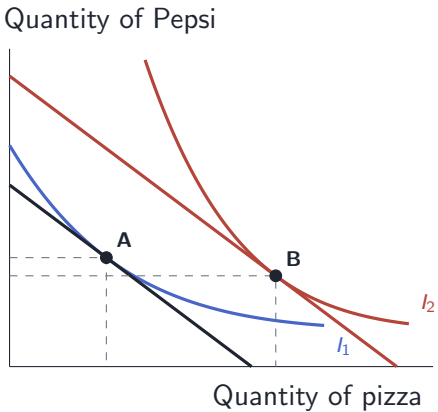


An increase in income: an inferior good

A consumer with different preferences, for whom Pepsi is an inferior good.

- Her income rises and the budget line shifts out in parallel.
- Her optimum moves from A to B, on the higher curve I_2.
- She buys more pizza (a normal good) and less Pepsi (an inferior good).

Finding. Whether a good is normal or inferior depends on preferences. The budget constraint alone does not decide it.

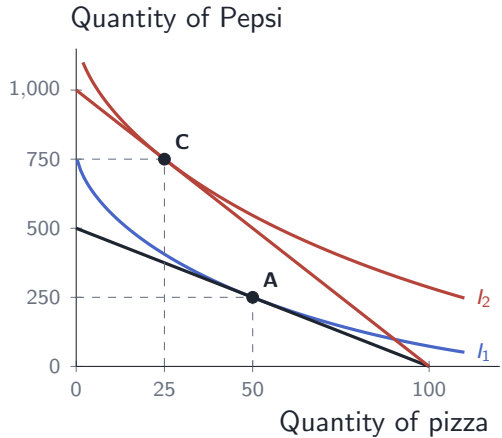


How changes in prices affect her choices

The price of Pepsi falls from \$2 to \$1. Income and the price of pizza do not change.

- The budget line pivots out around the pizza end point: 1,000 liters instead of 500.
- Her optimum moves from A $(50, 250)$ to C $(25, 750)$.
- She buys more Pepsi and less pizza.

Finding. At C, MRS $= 10/1 = 10$, the new relative price.



Income and substitution effects

A fall in the price of Pepsi has two effects.

Income effect. The change in consumption that results from the movement to a new indifference curve. Pepsi is cheaper, so her income buys more: she is in effect richer. If both goods are normal, she buys more of both.

Substitution effect. The change in consumption that results from moving to a new point on the same indifference curve, with a different MRS. Pizza now costs 10 liters instead of 5: pizza is relatively more expensive, so she buys less pizza and more Pepsi.

The two effects together

The price of Pepsi falls:

Good	Income effect	Substitution effect	Total effect
Pepsi	Richer: buys more Pepsi.	Pepsi relatively cheaper: buys more Pepsi.	Same direction: she buys more Pepsi.
Pizza	Richer: buys more pizza.	Pizza relatively dearer: buys less pizza.	Opposite directions: the effect on pizza depends on which is larger.

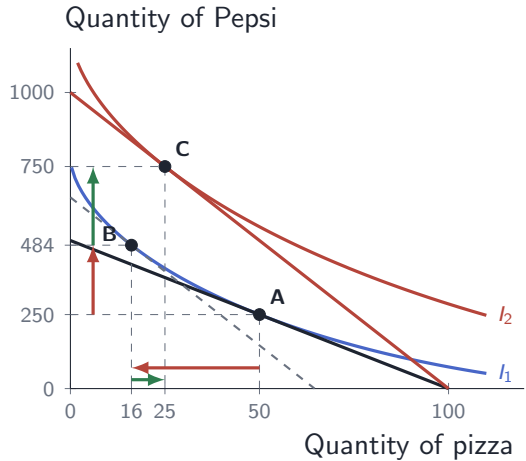
For the good whose price fell, the two effects agree. For the other good, they conflict.

Separating the two effects

Substitution effect, A to B: along I_1 to slope 10. Pizza 50 → 16, Pepsi 250 → 484.

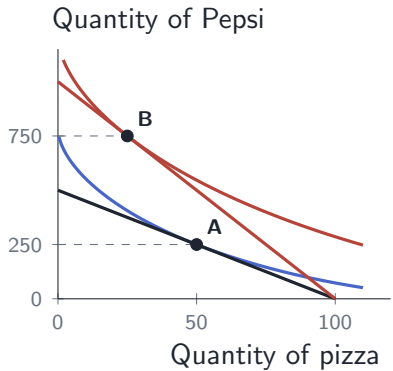
Income effect, B to C: out to I_2 at slope 10. Pizza 16 → 25, Pepsi 484 → 750.

Total, A to C: pizza 50 → 25, Pepsi 250 → 750.

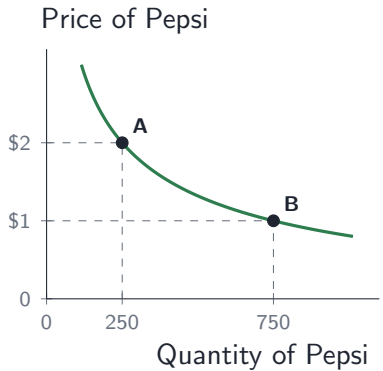


Red arrows: substitution effect. Green arrows: income effect.

Deriving the demand curve



(a) Her optimum. At \$2 a liter she buys 250 liters (A). At \$1 she buys 750 (B).



(b) Her demand curve for Pepsi. Each point is an optimum from panel (a).

Finding. A demand curve records the optimal choice at each price.

Part 4

Three Applications

Three applications

The same tools answer three questions about how the economy works.

1. Do all demand curves slope downward?
2. How do wages affect labor supply?
3. How do interest rates affect household saving?

Application 1: Do all demand curves slope downward?

The law of demand says that when a good's price rises, people buy less of it. In theory, a demand curve can slope upward.

A consumer buys meat and potatoes. Potatoes are a strongly inferior good: he buys many fewer when his income rises, and many more when it falls. The price of potatoes rises.

- **Substitution effect:** potatoes are relatively dearer, so he buys fewer potatoes and more meat.
- **Income effect:** he is poorer. Potatoes are inferior, so he buys more potatoes and less meat.

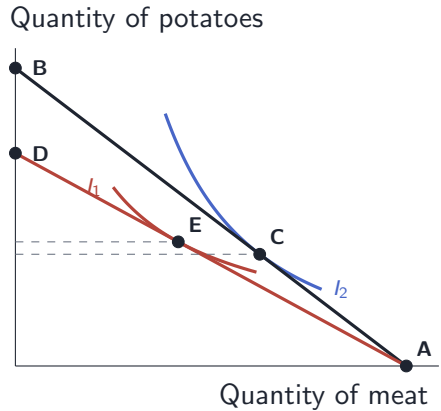
If the income effect is larger, he buys more potatoes at the higher price.

Giffen good. A good for which an increase in the price raises the quantity demanded. It is an inferior good whose income effect outweighs its substitution effect.

Application 1: A Giffen good

- At the low price of potatoes his budget line runs from A to B and his optimum is C.
- The price of potatoes rises. The line pivots in around A, to run from A to D.
- His new optimum is E, on the lower curve I_1. He is worse off, yet he buys **more** potatoes and less meat.

Finding. Potatoes are a Giffen good for this consumer.



Application 1: The search for Giffen goods

The Irish potato famine. Some historians suggest potatoes were a Giffen good in the 19th century. Potatoes were so large a part of the diet that a rise in their price cut living standards sharply. People cut back on meat and bought more of the staple, potatoes.

Rice in Hunan, China. Jensen and Miller (*American Economic Review*, 2008) gave randomly chosen poor households vouchers that subsidized rice. When the subsidy lowered the price of rice, many households ate *less* rice. When it was removed, they ate more.

Demand curves can slope upward, and sometimes do. Giffen goods are rare.

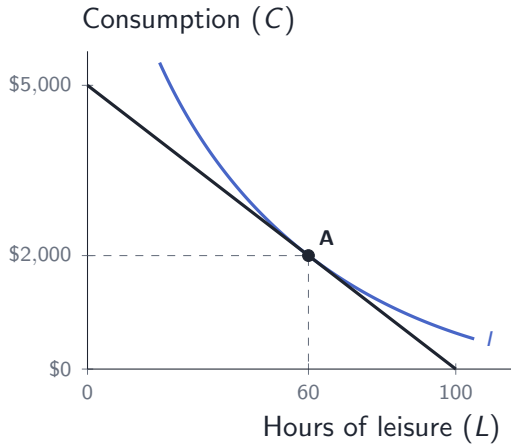
Application 2: How do wages affect labor supply?

A software designer is awake 100 hours a week. Each hour is either leisure or work, and each hour of work earns \$50 of consumption.

$$C = 50\,(100 - L)$$

- All leisure: $L = 100,\ C = 0$.
- All work: $L = 0,\ C = \$5{,}000$.
- Slope: each hour of leisure costs \$50 of consumption.

Finding. The wage is the price of leisure.



Application 2: Her optimum

Her optimum A is 60 hours of leisure, so she works 40 hours and consumes \$2,000. There, her MRS between leisure and consumption equals the wage, \$50.

- If her MRS were above \$50, an hour of leisure would be worth more to her than the \$50 it costs. She would gain by taking more leisure and working less.
- If her MRS were below \$50, an hour of leisure would be worth less than \$50. She would gain by working more.

At the optimum, MRS = wage: the same rule as MRS $= P_x/P_y$.

Application 2: An increase in the wage

Her wage rises from \$50 to \$60 an hour. All work now pays \$6,000, so the budget line pivots out and becomes steeper.

Substitution effect. Leisure is now more expensive relative to consumption: an hour costs \$60. She takes less leisure and works more.

Income effect. She is better off. If leisure is a normal good, she takes more leisure and works less.

The two effects pull hours of work in opposite directions. Which wins depends on her preferences.

Application 2: Case 1, the substitution effect is larger

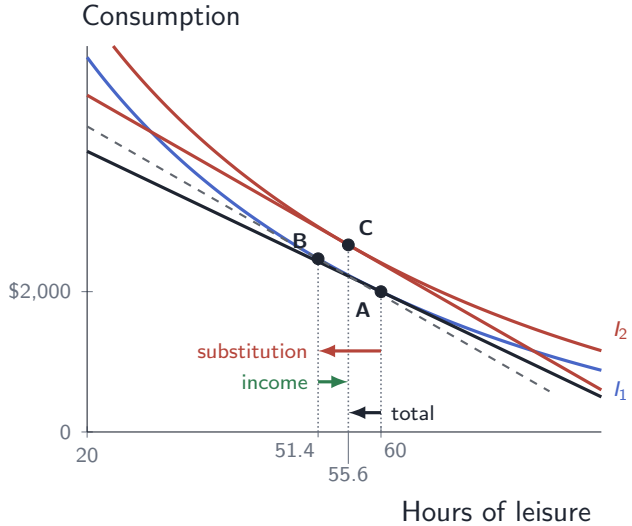
Wage \$50 → \$60. Optimum A → C.

Substitution, A to B: along I_1 to slope \$60. Leisure 60 → 51.4.

Income, B to C: out to the new line at the same slope. Leisure 51.4 → 55.6.

Total, A to C: −8.6 +4.2 = −4.4 hours of leisure. Work 40 → 44.4.

Finding. The substitution effect is larger.



Application 2: Case 2, the income effect is larger

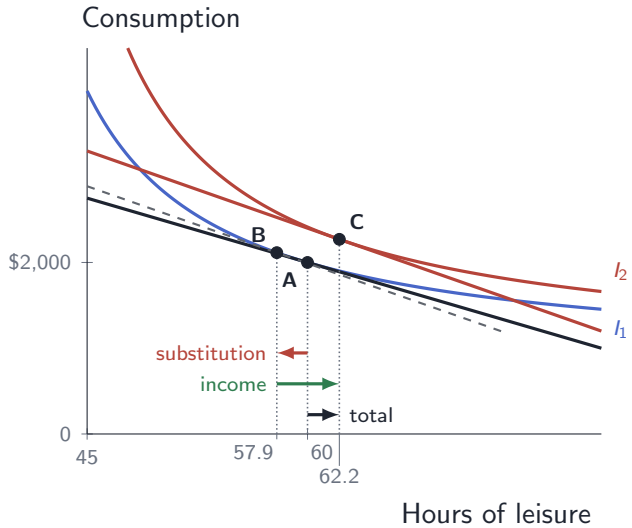
Wage \$50 → \$60. Optimum A → C.

Substitution, A to B: along I_1 to slope \$60. Leisure 60 → 57.9.

Income, B to C: out to the new line at the same slope. Leisure 57.9 → 62.2.

Total, A to C: $-2.1 + 4.3 = +2.2$ hours of leisure. Work 40 → 37.8.

Finding. The income effect is larger.

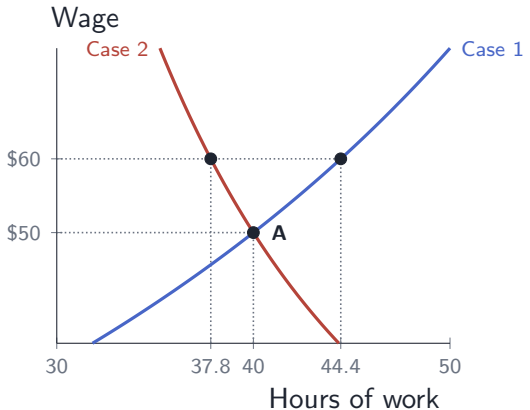


Application 2: Why the two cases differ

Hours of leisure	Case 1	Case 2
Substitution effect	−8.6	−2.1
Income effect	+4.2	+4.3
Total effect	−4.4	+2.2
Hours of work at \$60	44.4	37.8

The income effects are almost the same. The cases differ in the substitution effect, and that comes from the shape of her indifference curves.

In case 1 they bend gently: she readily swaps leisure for consumption, so a dearer hour of leisure moves her far along I_1. In case 2 they bend sharply: she is reluctant to swap, so the move is small.



Labor supply through A. Case 1 slopes upward; case 2 slopes backward.

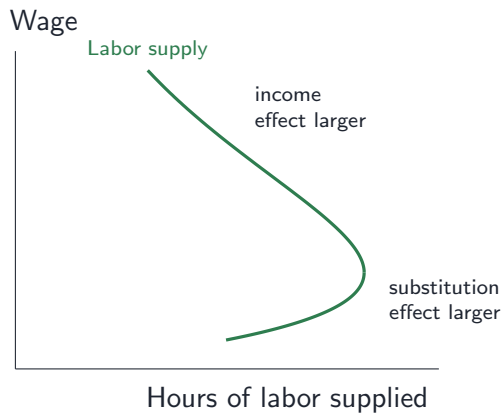
Application 2: A backward-bending labor-supply curve

The slope of labor supply need not be the same at every wage.

At low wages the substitution effect is larger: a higher wage brings more hours. The curve slopes upward.

At high wages the income effect is larger: a higher wage brings fewer hours. The curve bends backward.

Finding. Labor supply can slope upward at low wages and backward at high wages.



Application 2: The evidence

Historical trends. A hundred years ago many people worked six days a week; today five is the norm. Over the same period real wages have risen. Workers have used part of their higher prosperity to take more leisure: the income effect has dominated.

Lottery winners. A win raises income but leaves the wage unchanged, so there is no substitution effect: it isolates the income effect. A 2024 study in the *Quarterly Journal of Economics* found that for every dollar won, after-tax labor earnings fall by 40 cents. Winners near retirement age are much more likely to retire early.

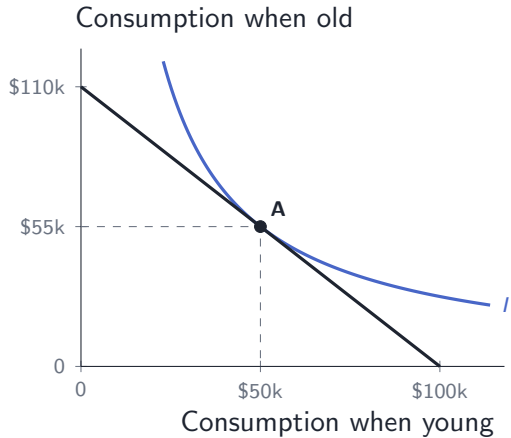
The income effect on labor supply is large.

Application 3: How do interest rates affect saving?

A person lives two periods. When young she earns \$100,000 and saves part of it. When old she consumes her saving plus interest. The interest rate is 10%.

$$C_{\text{old}} = 1.1 \times (100{,}000 - C_{\text{young}})$$

- Consume everything when young: \$100,000 then, nothing when old.
- Save everything: \$110,000 when old.
- Slope: each \$1 consumed young costs \$1.10 when old.



Application 3: Her optimum

At A she consumes \$50,000 when young and saves \$50,000. Saving earns 10%, so she consumes \$55,000 when old.

At A her MRS between consumption when young and consumption when old equals the relative price, \$1.10: one plus the interest rate.

The interest rate is the price of consuming now rather than later.

Application 3: An increase in the interest rate

The interest rate rises from 10% to 40%. Saving everything now gives \$140,000 when old, so the budget line pivots out and becomes steeper.

Substitution effect. Consuming when young is now more expensive: each \$1 costs \$1.40 of consumption when old. She consumes less when young and saves more.

Income effect. She is better off. If consumption in both periods is normal, she consumes more in both, so she consumes more when young and saves less.

The two effects pull saving in opposite directions.

Application 3: Case 1, the substitution effect is larger

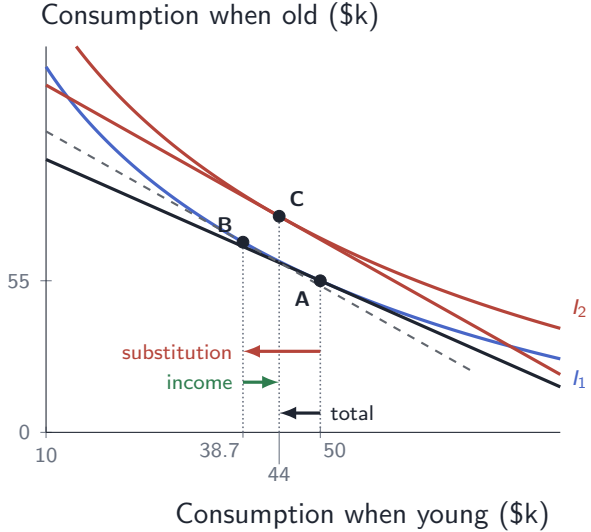
Interest 10% → 40%. Optimum A → C.

Substitution, A to B: along I_1 to slope 1.40. Young \$50k → \$38.7k.

Income, B to C: out to the new line. Young \$38.7k → \$44.0k.

Total, A to C: −\$11.3k +\$5.3k = −\$6.0k. Saving \$50k → \$56.0k.

Finding. The substitution effect is larger.



Application 3: Case 2, the income effect is larger

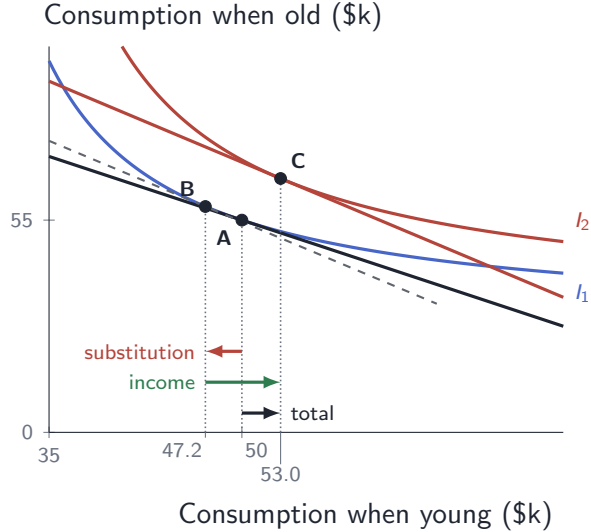
Interest 10% → 40%. Optimum A → C.

Substitution, A to B: along I_1 to slope 1.40. Young \$50k → \$47.2k.

Income, B to C: out to the new line. Young \$47.2k → \$53.0k.

Total, A to C: −\$2.8k +\$5.8k = +\$3.0k. Saving \$50k → \$47.0k.

Finding. The income effect is larger.



Application 3: What it means for policy

A tax on interest income lowers the return on saving, in the same way as a fall in the interest rate.

- The substitution effect lowers saving: saving pays less.
- The income effect raises saving: she is poorer, so she consumes less when young.

Theory alone cannot say whether taxing interest income reduces saving. The answer depends on which effect is larger.

Part 5

Conclusion

Do people really think this way?

Consumers do not draw indifference curves or compute marginal rates of substitution before they shop.

The theory does not claim that they do. It describes the outcome of their choices: people do the best they can with what they can afford. That is enough to explain why demand curves usually slope down, and when labor supply and saving can respond either way.

The model is a description of choices, not of the thought process behind them.

Chapter summary (1)

- The **budget constraint** $P_x x + P_y y = M$ shows the bundles a consumer can afford. Its end points are M/P_x and M/P_y. Its slope is the relative price, P_x/P_y.
- **Indifference curves** represent preferences. Higher curves are preferred. The slope of an indifference curve is the marginal rate of substitution.
- **Marginal utility** is the slope of the total-utility curve. The MRS equals MU_x/MU_y.

Chapter summary (2)

- The consumer **optimizes** at the point on her budget constraint that reaches the highest indifference curve. There, MRS $= P_x/P_y$, or equivalently $MU_x/P_x = MU_y/P_y$.
- A price change has an **income effect** (a move to a new indifference curve) and a **substitution effect** (a move along the original curve to a new MRS). A demand curve records the optimum at each price.
- The theory explains why demand curves can slope upward, why higher wages can raise or lower labor supply, and why higher interest rates can raise or lower saving.